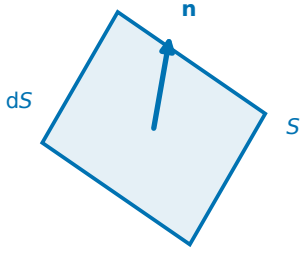
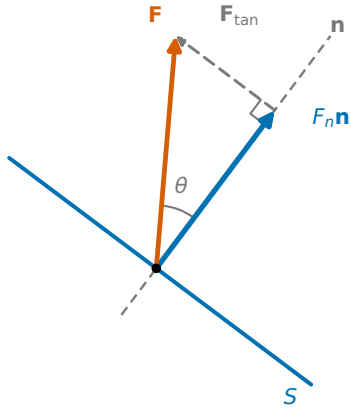
A



$$\mathrm{d}\mathbf{S} = \mathbf{n}\,\mathrm{d}S$$

B



$$F_n = \mathbf{F}\cdot\mathbf{n} = \|\mathbf{F}\|\cos\theta$$

$$\mathbf{F}\cdot\mathrm{d}\mathbf{S} = (\mathbf{F}\cdot\mathbf{n})\,\mathrm{d}S = F_n\,\mathrm{d}S$$